

File permissions in Linux

Project description

The organization's research team needs to update permissions for specific files and directories within the `projects` directory. The current permissions do not align with the privilege levels the users should have. Performing a security check on permissions will help maintain system security. To carry out this inspection, I performed the following tasks:

Check file and directory details

The commands below illustrate how to navigate to the `projects` directory and view the permissions for its files and sub-directories:

```
researcher2@d18750a850d6:~$ cd projects
researcher2@d18750a850d6:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Oc
t  2 16:16 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Oc
t  2 16:16 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Oc
t  2 16:16 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oc
t  2 16:16 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oc
t  2 16:16 project_t.txt
researcher2@d18750a850d6:~/projects$
```

Describe the permissions string

The 10-character string can be broken down as follows:

- **1st character:** indicates the file type. "-" for a file and "d" for a directory.
- **2nd–4th characters:** Indicate permissions for the user: "r" for read, "w" for write, "x" for execute, and "-" to indicate the user lacks a specific permission.
- **5th–7th characters:** Indicate permissions for the group: "r" for read, "w" for write, "x" for execute, and "-" to indicate the group lacks a specific permission.
- **8th–10th characters:** Indicate permissions for others (users outside the group who are not the owner): "r" for read, "w" for write, "x" for execute, and "-" to indicate that others lack a specific permission.

Change file permissions

The organization determined that "others" should not have write permission. Therefore, I removed write permission for "others" from the `project\_k.txt` file:

```
researcher2@d18750a850d6:~/projects$ chmod o-w
project_k.txt
researcher2@d18750a850d6:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oc
t  2 16:16 .
drwxr-xr-x 3 researcher2 research_team 4096 Oc
t  2 17:41 ..
-rw--w---- 1 researcher2 research_team  46 Oc
t  2 16:16 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oc
t  2 16:16 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Oc
t  2 16:16 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Oc
t  2 16:16 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oc
t  2 16:16 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oc
t  2 16:16 project_t.txt
researcher2@d18750a850d6:~/projects$
```

Change file permissions on a hidden file

The organization determined anyone should not have write permission. Therefore, I removed write permission for all from the `.project\_x.txt` file:

```
researcher2@d18750a850d6:~/projects$ chmod u-w
,g-w,g+r .project_x.txt
researcher2@d18750a850d6:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oc
t  2 16:16 .
drwxr-xr-x 3 researcher2 research_team 4096 Oc
t  2 17:41 ..
-r--r----- 1 researcher2 research_team  46 Oc
t  2 16:16 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Oc
t  2 16:16 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Oc
t  2 16:16 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Oc
t  2 16:16 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oc
t  2 16:16 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oc
t  2 16:16 project_t.txt
researcher2@d18750a850d6:~/projects$
```

Change directory permissions:

My organization wants only the user `researcher2` to have execute permission for the `drafts` subdirectory:

```
researcher2@d18750a850d6:~/projects$ chmod g-x  
drafts/  
researcher2@d18750a850d6:~/projects$ ls -la  
total 32  
drwxr-xr-x 3 researcher2 research_team 4096 Oc  
t  2 16:16 .  
drwxr-xr-x 3 researcher2 research_team 4096 Oc  
t  2 17:41 ..  
-rw--w---- 1 researcher2 research_team  46 Oc  
t  2 16:16 .project_x.txt  
drwx----- 2 researcher2 research_team 4096 Oc  
t  2 16:16 drafts  
-rw-rw-r-- 1 researcher2 research_team  46 Oc  
t  2 16:16 project_k.txt  
-rw-r----- 1 researcher2 research_team  46 Oc  
t  2 16:16 project_m.txt  
-rw-rw-r-- 1 researcher2 research_team  46 Oc  
t  2 16:16 project_r.txt  
-rw-rw-r-- 1 researcher2 research_team  46 Oc  
t  2 16:16 project_t.txt  
researcher2@d18750a850d6:~/projects$
```

Summary

changed multiple permissions to match the level of authorization my organization wanted for files and directories in the projects directory. The first step in this was using `ls -la` to check the permissions for the directory. This informed my decisions in the following steps. I then used the `chmod` command multiple times to change the permissions on files and directories.